

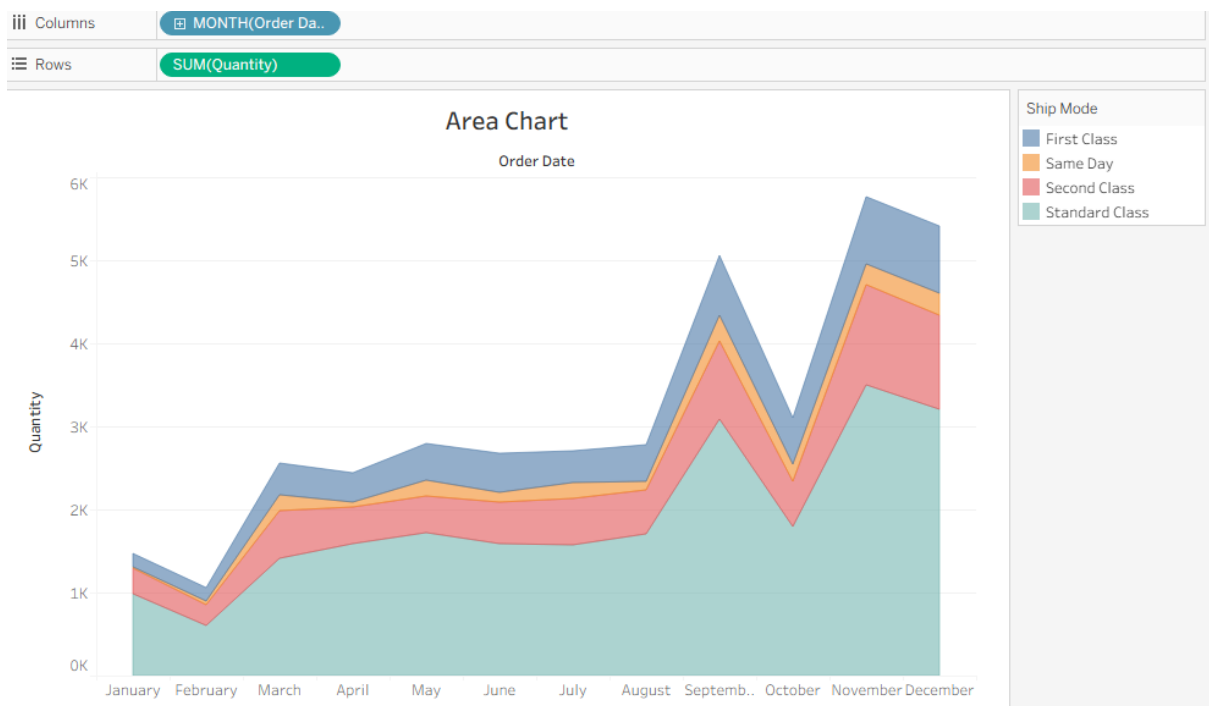
|                     |                            |
|---------------------|----------------------------|
| Hall Ticket         | 0425046036                 |
| Of Name the Student | SAMADHI HARSHITHA          |
| Learning Track      | DATA ANALYTIC WITH TABLEAU |

## Assignment

### Area Chart:

1. First, I took Order Date and dropped it in the column box.
2. Then I right click year (Order Date) in the columns.
3. I click on month to see monthly data.
4. Then I dragged Quantity and put in the side row box.
5. To add color, I took ship mode dragged it color in the marks box.
6. In the marks box, I clicked to see chart options.
7. Select the area chart in chart type.

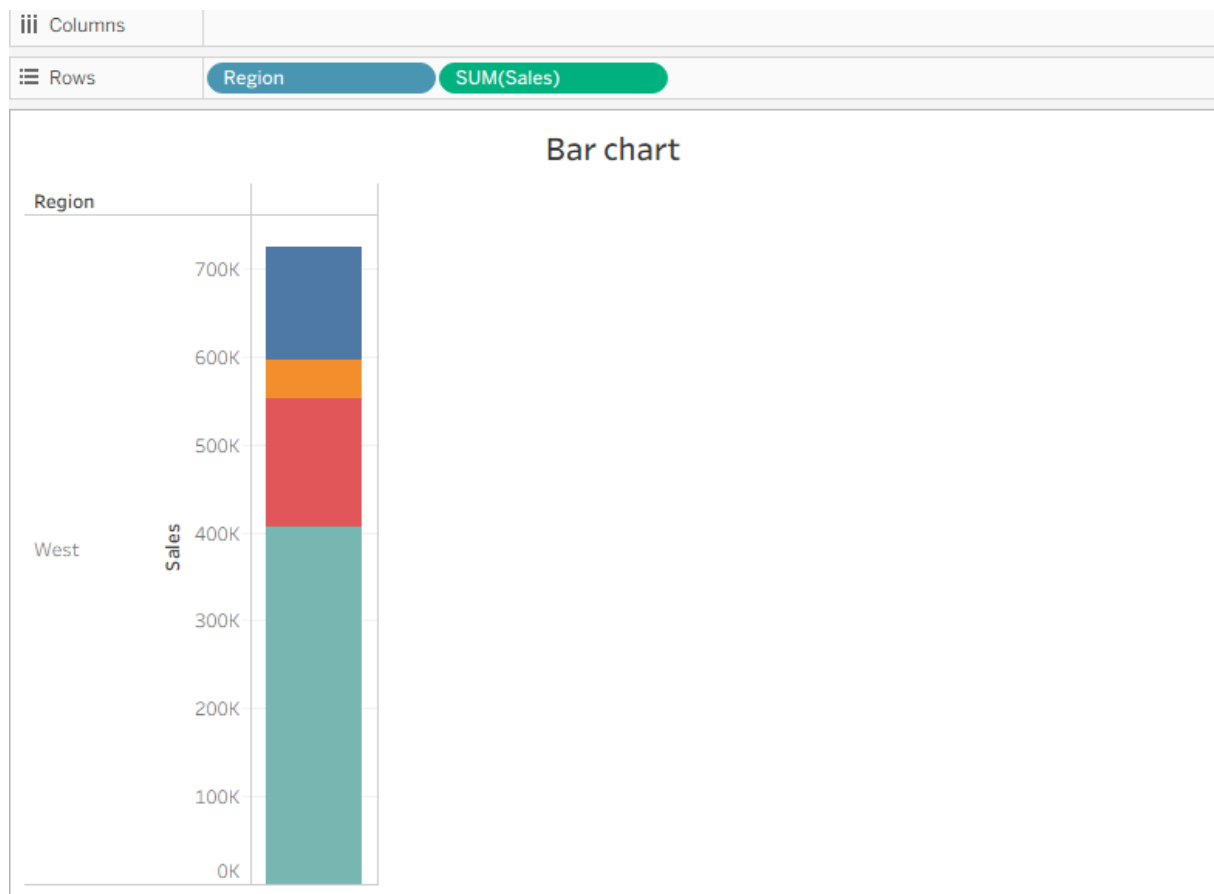
### OUTPUT:



### Bar Chart:

1. First, I dragged sales and dropped it in the Rows box.
2. In marks, I changed the chart type to bar.
3. I dragged ship mode to color for different segments.
4. Then I added region to rows to compare my region.
5. I filtered region and selected only west remaining unchecked.
6. Click ok.

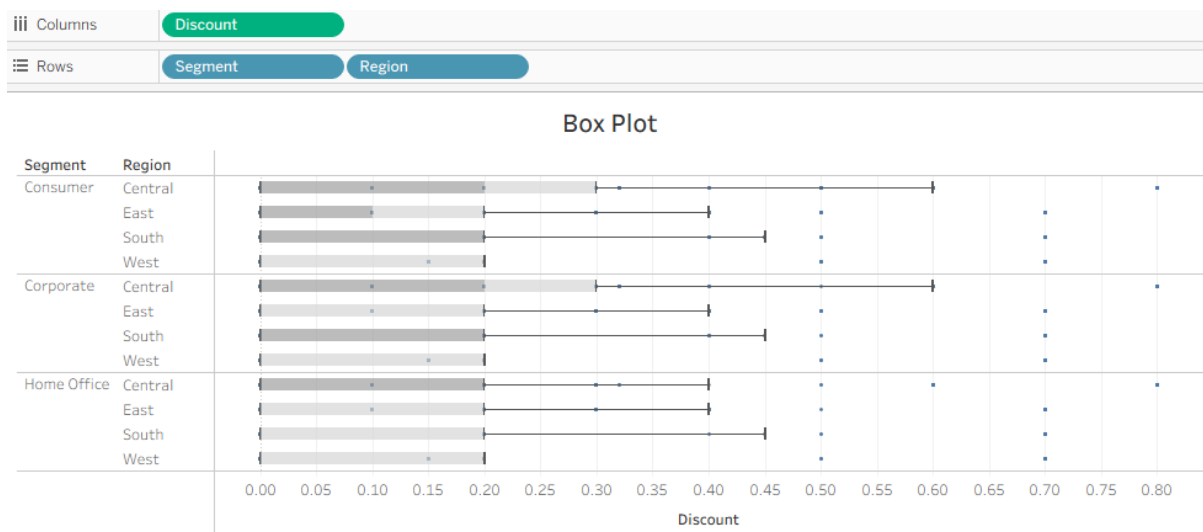
## OUTPUT:



## Box plot:

1. First, I dragged segment to the columns shelf.
2. Then I dragged discount on the rows shelf.
3. I right clicked Discount on rows and hovered on measure.
4. From there, I changed the aggregation to average.
5. Next, I added region to columns shelf also.
6. Region got placed on the right side of segment
7. I clicked show me from the toolbar at the top.
8. Then I selected box and whisker plot from show me.
9. I moved region next to segment in columns for better view.
10. I went to analysis menu and unchecked aggregate measures.
11. Finally, I clicked swap rows and columns button to adjust layout

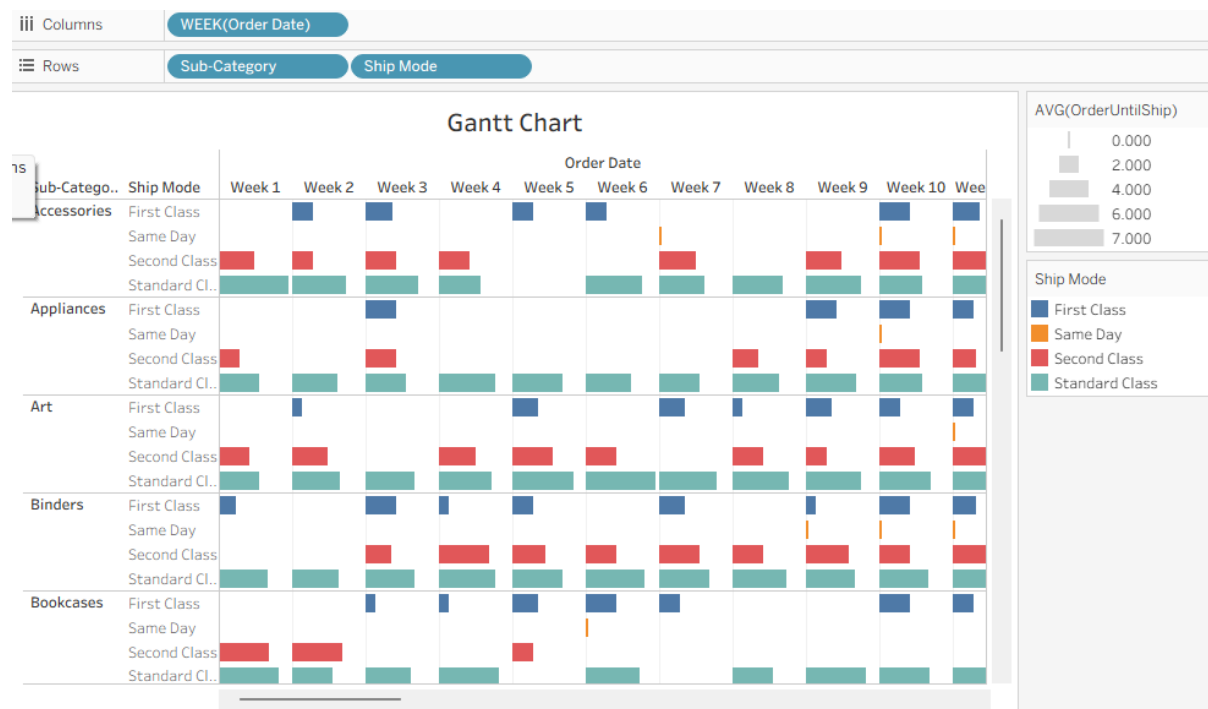
## Output:



## Gantt chart:

1. I dragged order date to columns shelf.
2. I clicked the dropdown on YEAR (Order date) and changed it to week numbers.
3. Then I put sub-category in rows shelf.
4. I added ship mode also in rows.
5. In marks card, I changed chart type to Gantt bar.
6. For shipping delay, I made a calculated field named OrderUntilShip.
7. Formula used: DATEDIFF ('day', [order date], [ship date]).
8. I dragged OrderUntilShip to size.
9. I right-clicked SUM(OrderuntilShip) and changed measure to average.
10. Finally, I dragged ship mode to color for different colours.

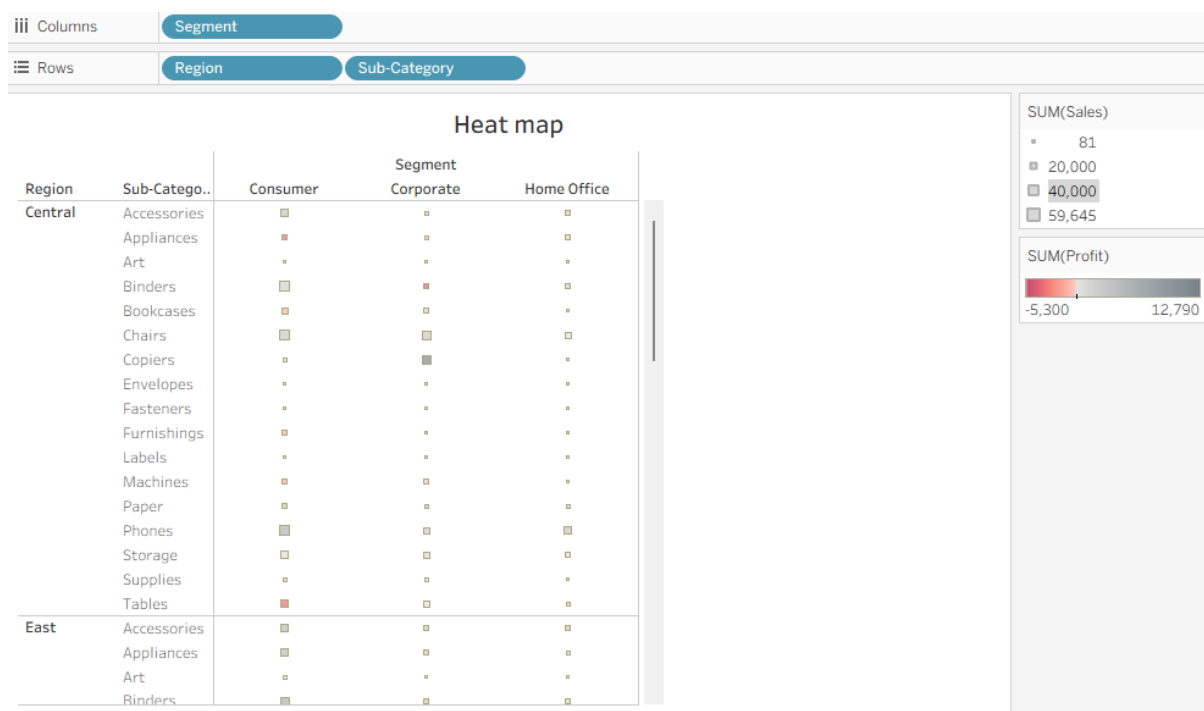
## Output:



## Heat Map:

1. I dragged segment to the columns shelf. It shows consumer, corporate, home office.
2. Then I put Region in the Rows shelf.
3. Next to Region. I added sub-category also in Rows.
4. I dragged profit to color in marks card.
5. Now blue boxes = high profit, profit, orange boxes = loss
6. I clicked color in marks and set Border to Medium Gray for clear boxes.
7. Then I clicked the profit color legend selected Edit colours.
8. I chose Red- Black diverging palette and ticked use full color Range.
9. After Apply black= profit, red=loss.
10. Initially, I dragged sales to size in marks card.
11. Now bigger boxes= high sales, smaller boxes= low sales.

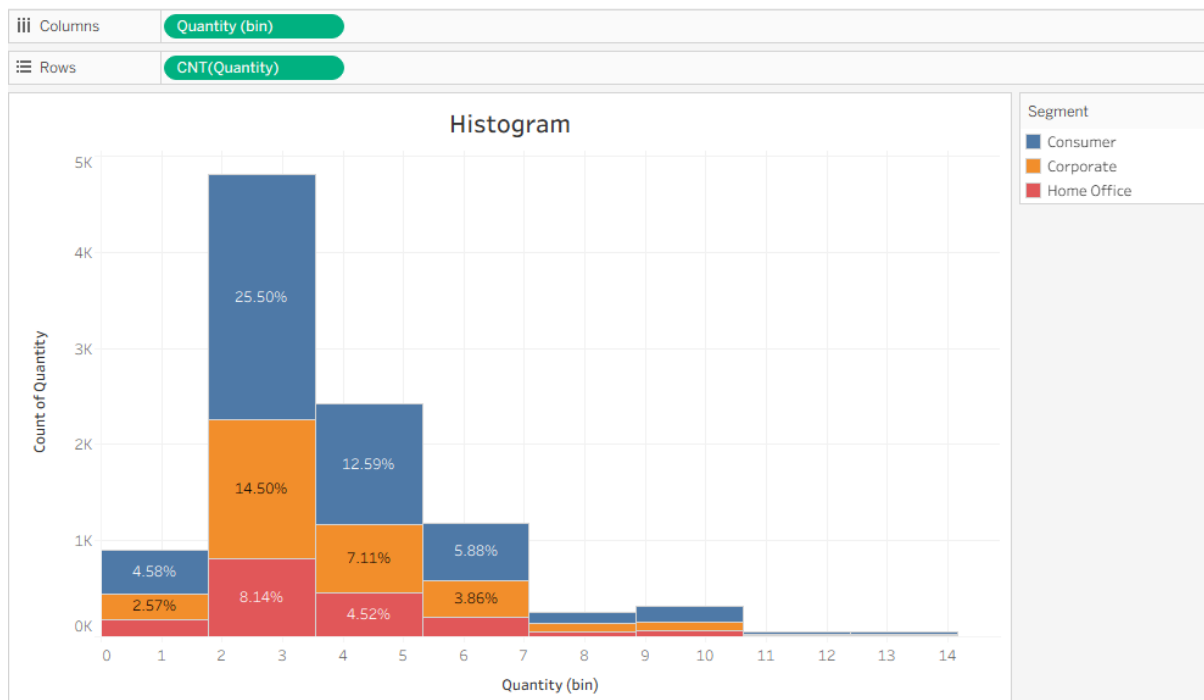
## Output:



## Histogram:

1. I dragged Quantity from Measures to Columns shelf.
2. I clicked Show Me and selected Histogram.
3. Tableau created Quantity (bin) on Columns and CNT(Quantity) on Rows.
4. X-Axis shows Quantity Bins; Y-Axis shows Count of Orders.
5. I dragged Segment to Colour in Marks card to see Consumer, Corporate, Home Office.
6. I held Ctrl and dragged CNT(Quantity) to Label.
7. I right-clicked CNT(Quantity) on Marks card and selected Quick Table Calculation → Percent of Total.
8. Key Insight: Most orders have quantity 2 or 3. Consumer segment has the highest orders.

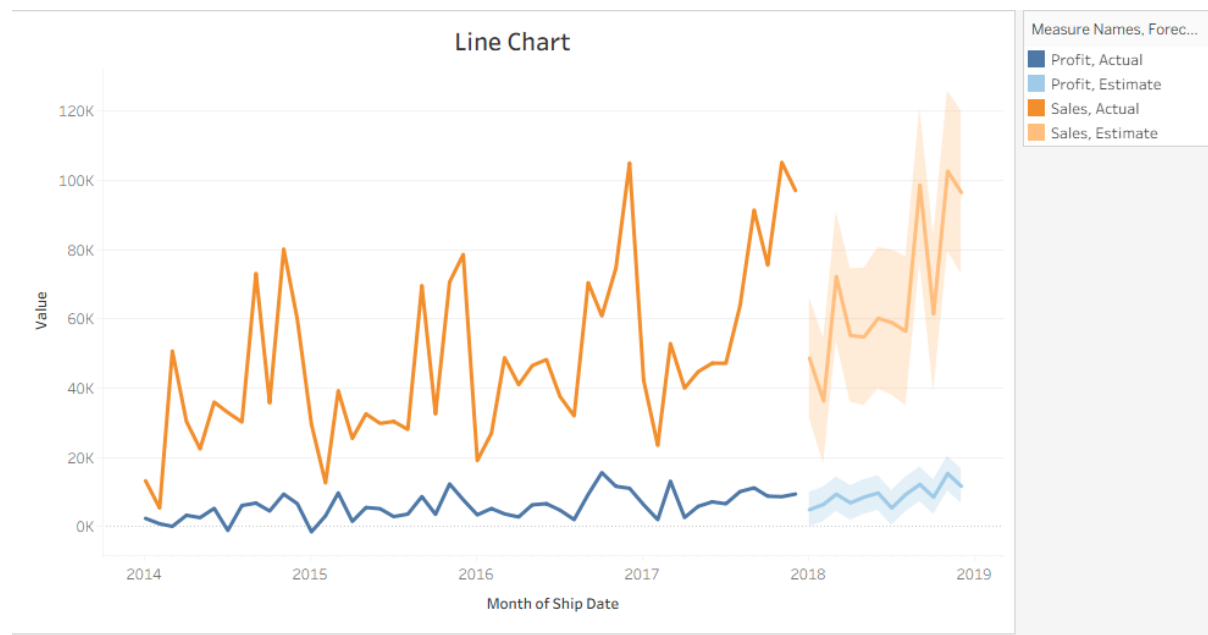
## Output:



## Line chart:

1. I dragged Ship Date to Columns. Tableau showed years.
2. I dragged Sales to Rows. A line chart came up.
3. I dragged Profit to Rows next to Sales. It made two separate axes.
4. I dragged Profit and dropped it on the Sales Axis to combine them. Now both use one axis.
5. I clicked the dropdown on YEAR(Ship Date) and changed it to Month. Now it shows monthly trend for all years.
6. Key Insight: Sales and Profit go up near year-end. This pattern happens every year. It is called Seasonality.
7. I went to the Analytics Pane and dragged Forecast to the chart. Tableau showed the future trend.

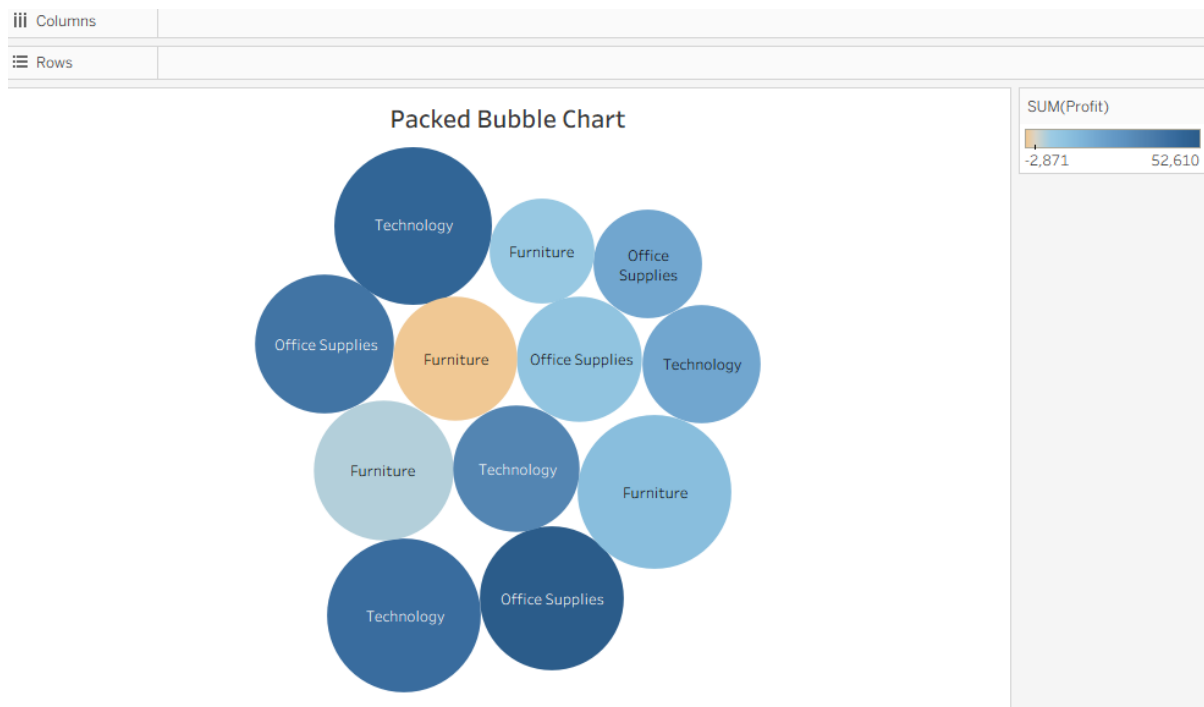
## Output:



## Bubble chart:

1. I dragged Category from Dimensions to Columns. I saw Furniture, Office Supplies, and Technology.
2. I dragged Sales from Measures to Rows. Tableau made a bar chart first.
3. I clicked Show Me and selected Packed Bubbles. Now I got big and small bubbles for each category. Bubble size shows Sales.
4. I dragged Region to Detail in the Marks card. Each category split into East, West, Central, and South bubbles.
5. I dragged Profit to Colour. Dark blue bubbles mean high profit. Light blue means low profit. Orange/Red means loss.
6. I dragged Region to Label. Now each bubble shows the region name.
7. Key Insight: Technology has the biggest bubbles, so highest sales. Some bubbles are red, which means loss in that region and category.

## Output:

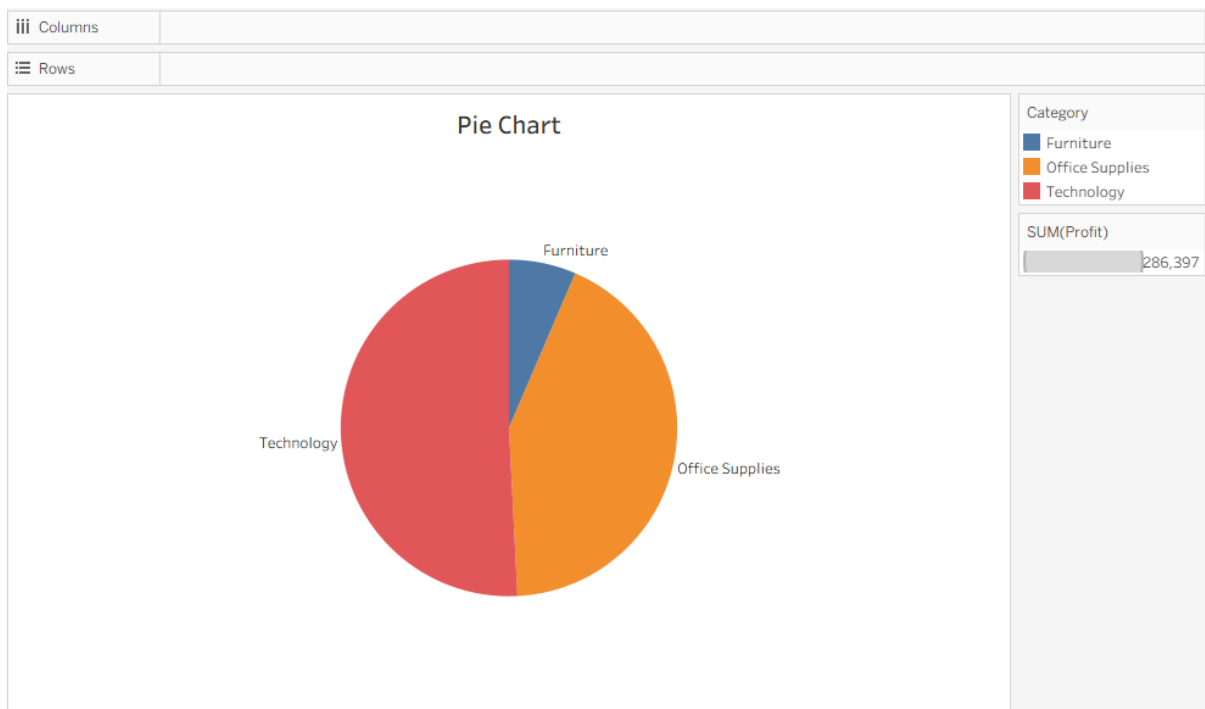


## Pie chart:

1. I dragged Profit from Measures to Columns.
2. I dragged Category from Dimensions to Rows. Tableau first showed a bar chart.
3. I clicked Show Me and selected Pie Chart. It changed to a pie automatically.
4. Tableau put Profit on Angle and Size, and Category on Colour. So, each slice is a category.
5. I made the pie bigger by clicking Fit → Entire View or by increasing Size in the Marks card.
6. I dragged Category to Label so each slice shows Furniture, Office Supplies, and Technology.
7. Key Insight: The biggest slice shows which category has the highest profit. It helps compare profit share easily.



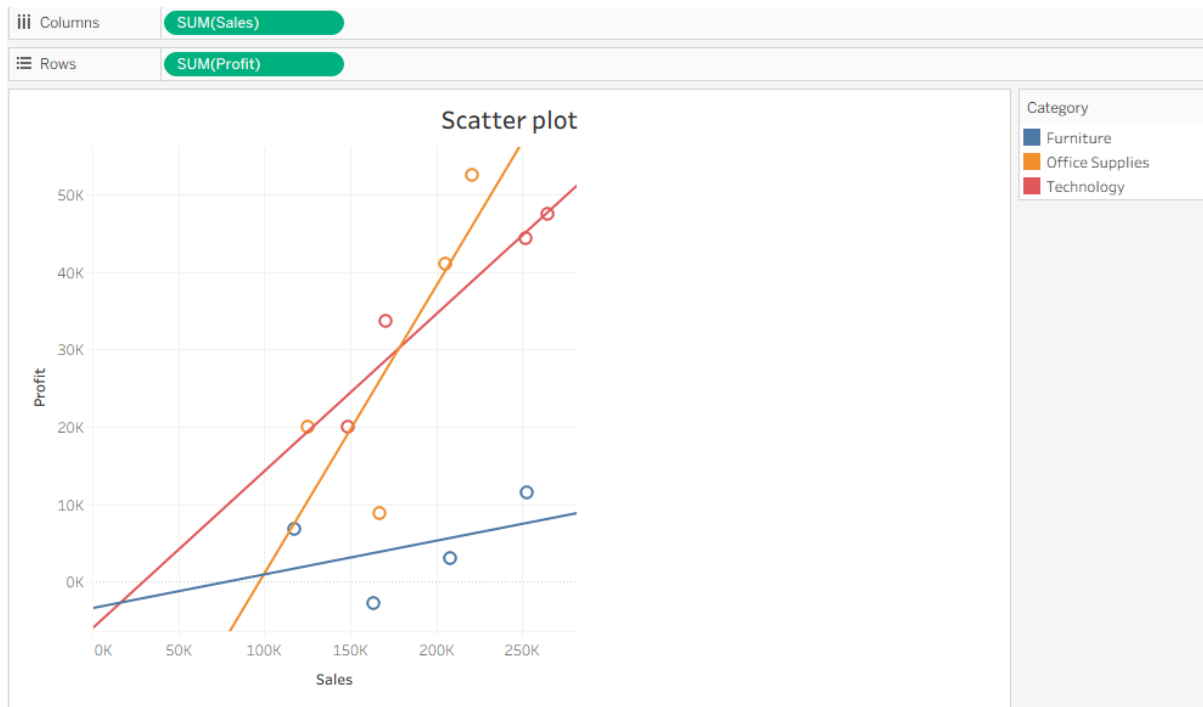
## Output:



## Scatter plot:

1. I put Sales on Columns.
2. I put Profit on Rows. A scatter plot appeared.
3. I added Category to Colour. Dots split by Furniture, Office Supplies, Technology.
4. I added Region to Detail. More dots came for Central, East, South, West.
5. I dragged Trend Line from Analytics Pane and used Linear.
6. Key Insight: Line going up means more sales = more profit. Dots below line = loss.

## Output:



## Text Table with Percentage:

1. I put Order Date on Columns. It showed years.
2. I put Sub-Category on Rows. A table came up.
3. I put Sales on Text. Now cells show sales numbers.
4. I put Region on Rows before Sub-Category. Table split by region → Sub-Category.
5. I right-clicked SUM(Sales) → Add Table Calculation → Percent of Total.
6. I set Compute Using to Pane (Down) and clicked OK. Numbers became %.
7. Key Insight: This shows which region and sub-category got what % of sales each year.

## Output:

|         |                     |
|---------|---------------------|
| Columns | YEAR(Order Date)    |
| Rows    | Region Sub-Category |

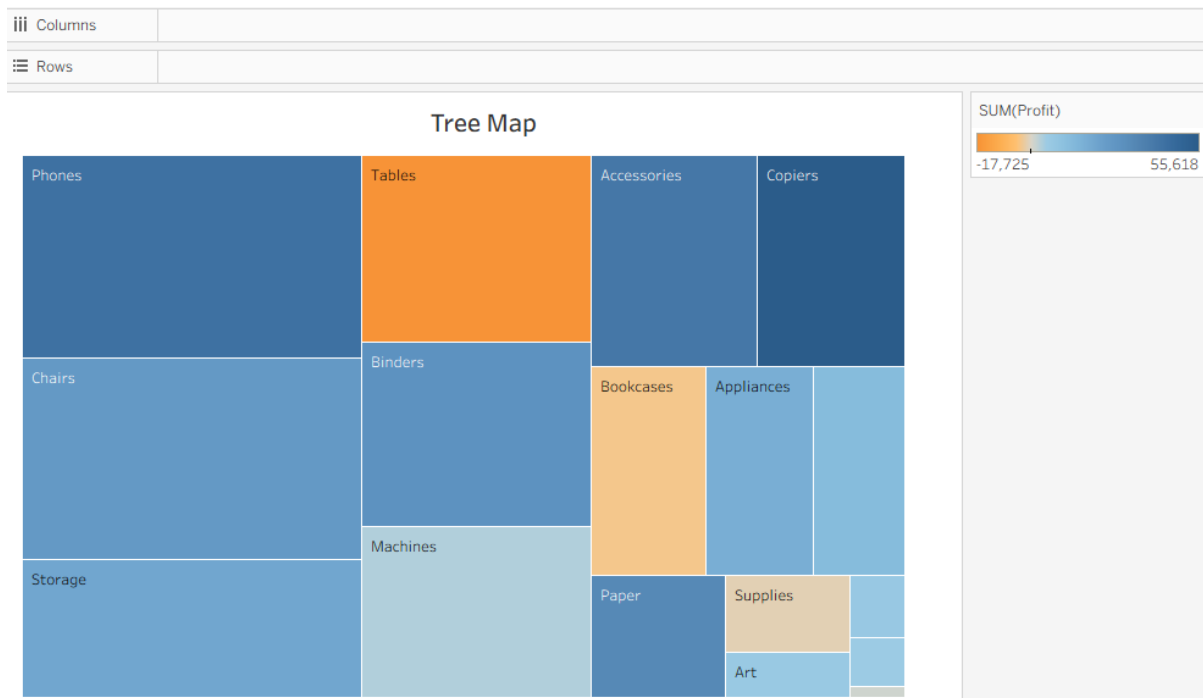
Text Table with Percentage

| Region  | Sub-Catego.. | Order Date |        |        |        |
|---------|--------------|------------|--------|--------|--------|
|         |              | 2014       | 2015   | 2016   | 2017   |
| Central | Accessories  | 4.27%      | 7.58%  | 7.33%  | 7.42%  |
|         | Appliances   | 3.52%      | 4.84%  | 4.08%  | 6.07%  |
|         | Art          | 0.79%      | 1.10%  | 1.03%  | 1.56%  |
|         | Binders      | 15.28%     | 5.73%  | 9.53%  | 14.35% |
|         | Bookcases    | 1.77%      | 8.07%  | 5.69%  | 3.83%  |
|         | Chairs       | 19.99%     | 17.41% | 15.84% | 15.78% |
|         | Copiers      | 3.15%      | 12.45% | 11.87% | 2.50%  |
|         | Envelopes    | 1.54%      | 0.85%  | 0.66%  | 0.81%  |
|         | Fasteners    | 0.12%      | 0.09%  | 0.17%  | 0.22%  |
|         | Furnishings  | 2.44%      | 2.46%  | 3.47%  | 3.45%  |
|         | Labels       | 1.01%      | 0.30%  | 0.35%  | 0.40%  |
|         | Machines     | 15.69%     | 1.80%  | 1.80%  | 4.08%  |
|         | Paper        | 2.26%      | 3.45%  | 3.64%  | 4.24%  |
|         | Phones       | 9.56%      | 18.82% | 13.50% | 15.78% |
|         | Storage      | 10.68%     | 8.10%  | 8.69%  | 9.31%  |
|         | Supplies     | 0.42%      | 0.32%  | 2.91%  | 3.00%  |
|         | Tables       | 7.50%      | 6.67%  | 9.44%  | 7.20%  |
| East    | Accessories  | 4.70%      | 11.46% | 3.45%  | 6.96%  |
|         | Appliances   | 4.49%      | 4.28%  | 5.22%  | 5.77%  |
|         | Art          | 1.00%      | 1.09%  | 1.04%  | 1.22%  |
|         | Binders      | 4.93%      | 9.09%  | 10.49% | 6.57%  |

## Tree map:

1. I put Sub-Category on Columns. I saw sub-category names.
2. I put Sales on Rows. Tableau first made a bar chart.
3. I clicked Show Me and selected Tree map. Now it changed to rectangles.
4. In tree map, big rectangle = high sales. Small rectangle = low sales.
5. I dragged Ship Mode to Colour. Now rectangles got different colours for First Class, Second Class, Standard Class, Same Day.
6. If I want profit, I remove Ship Mode and drag Profit to Colour instead. Then dark colour = high profit, light colour = low profit. Key Insight: Biggest box shows which sub-category has highest sales. Colour shows ship mode or profit.

## Output:

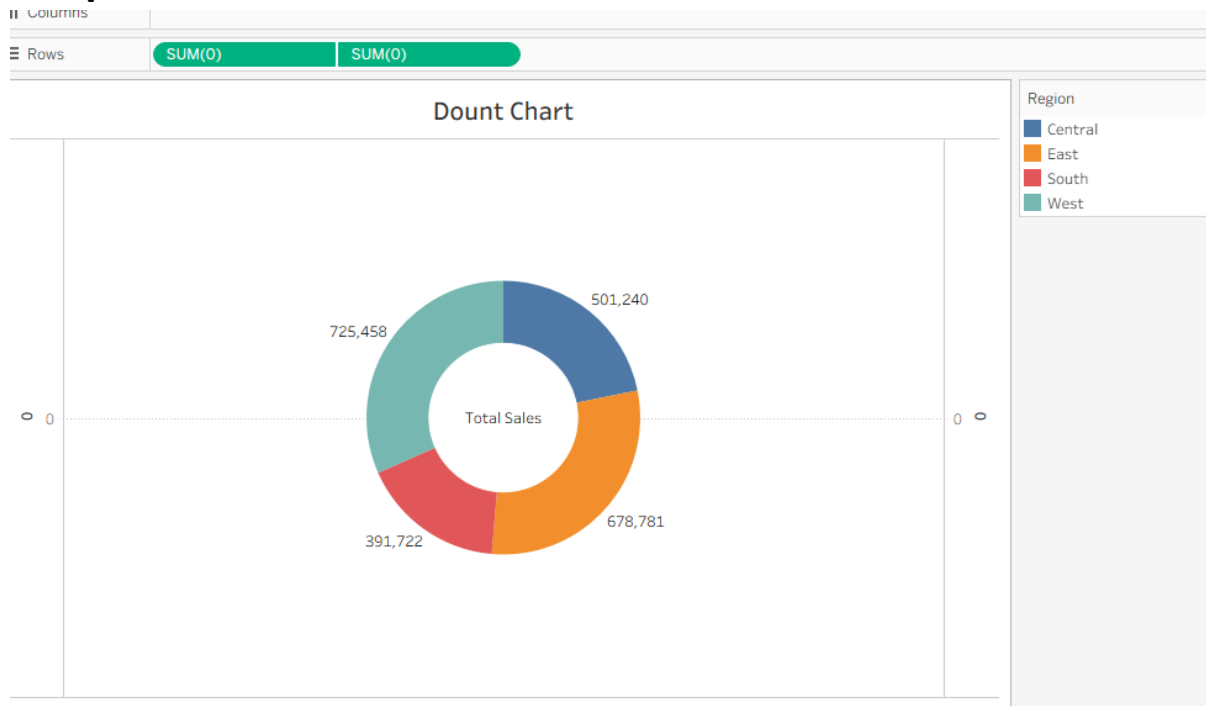


## Donut chart:

1. I went to Analysis and clicked Create Calculated Field. I named it Zero and typed 0. Then clicked OK.
2. I dragged Zero to Rows. It made a blank axis.
3. In Marks card I changed Automatic to Pie. I clicked Fit → Entire View. A big circle came.
4. I increased the Size slider to make the pie bigger.
5. I dragged Zero to Rows again, next to the first one. Now I got two circles.
6. I right-clicked the second SUM(Zero) and clicked Dual Axis. Both circles overlapped.
7. I clicked the second SUM(Zero) Marks card. I made Size smaller and set Colour to White. This made the donut hole.
8. I clicked the first SUM(Zero) Marks card. I dragged Region to Colour. Each region got a different colour.
9. I dragged Region to Label, Sales to Label, and Sales to Angle. Now slices show region and sales.
10. I clicked the second SUM(Zero) Marks card. I dragged Sales to Label. I edited Label text and typed Total Sales. Now the centre shows total sales.

11. Key Insight: Donut chart shows sales share by region. The hole in center shows total sales. Looks cleaner than pie chart.

## Output:

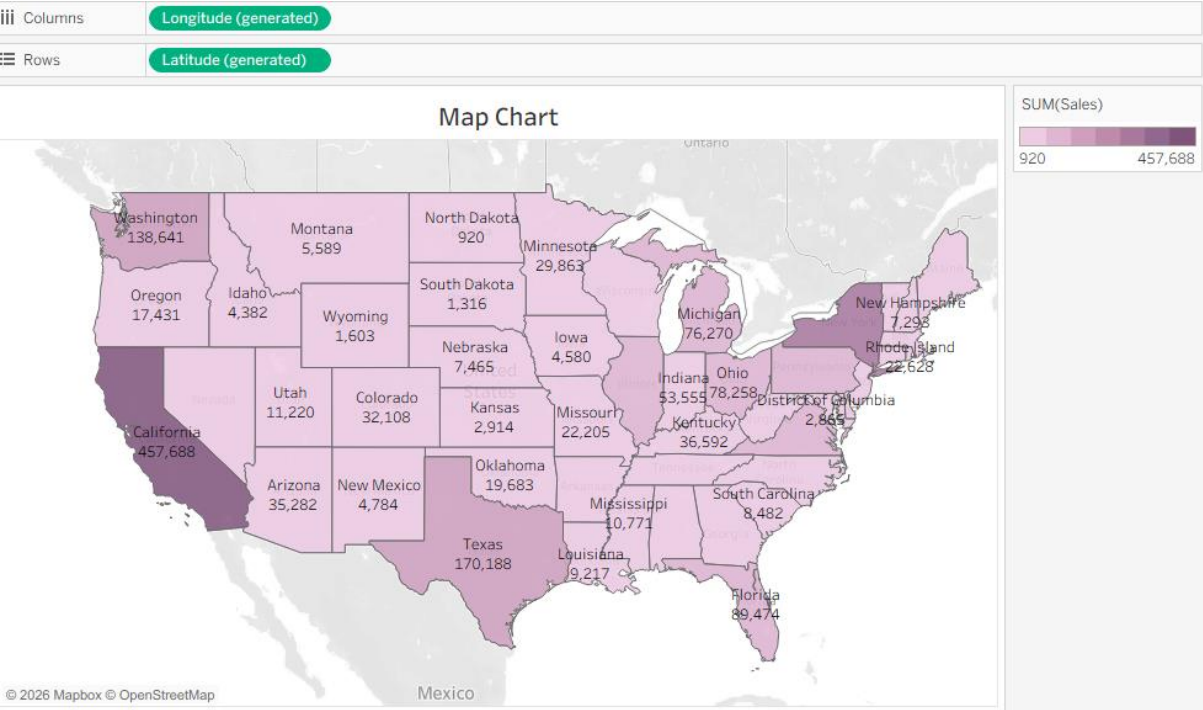


## Map chart:

1. I dragged State to the workspace. Tableau made a map automatically.
2. I dragged Sales to Colour in Marks card. Now states got different colours based on sales. Dark = high sales, light = low sales.
3. In Marks card I changed Automatic to Filled Map. Now full states are coloured, not just dots.
4. I clicked Colour → Edit Coors and picked a Purple palette. Then clicked OK.
5. I dragged State to Label and Sales to Label. Now each state shows its name and sales value.
6. I dragged Region to Filters and selected All regions. Now I can filter by region.
7. I dragged Profit to Tooltip. When I hover on a state, it shows sales and profit.
8. I went to Map → Map Layers. I set Background style to Outdoors and Washout to 70% to make it clean.

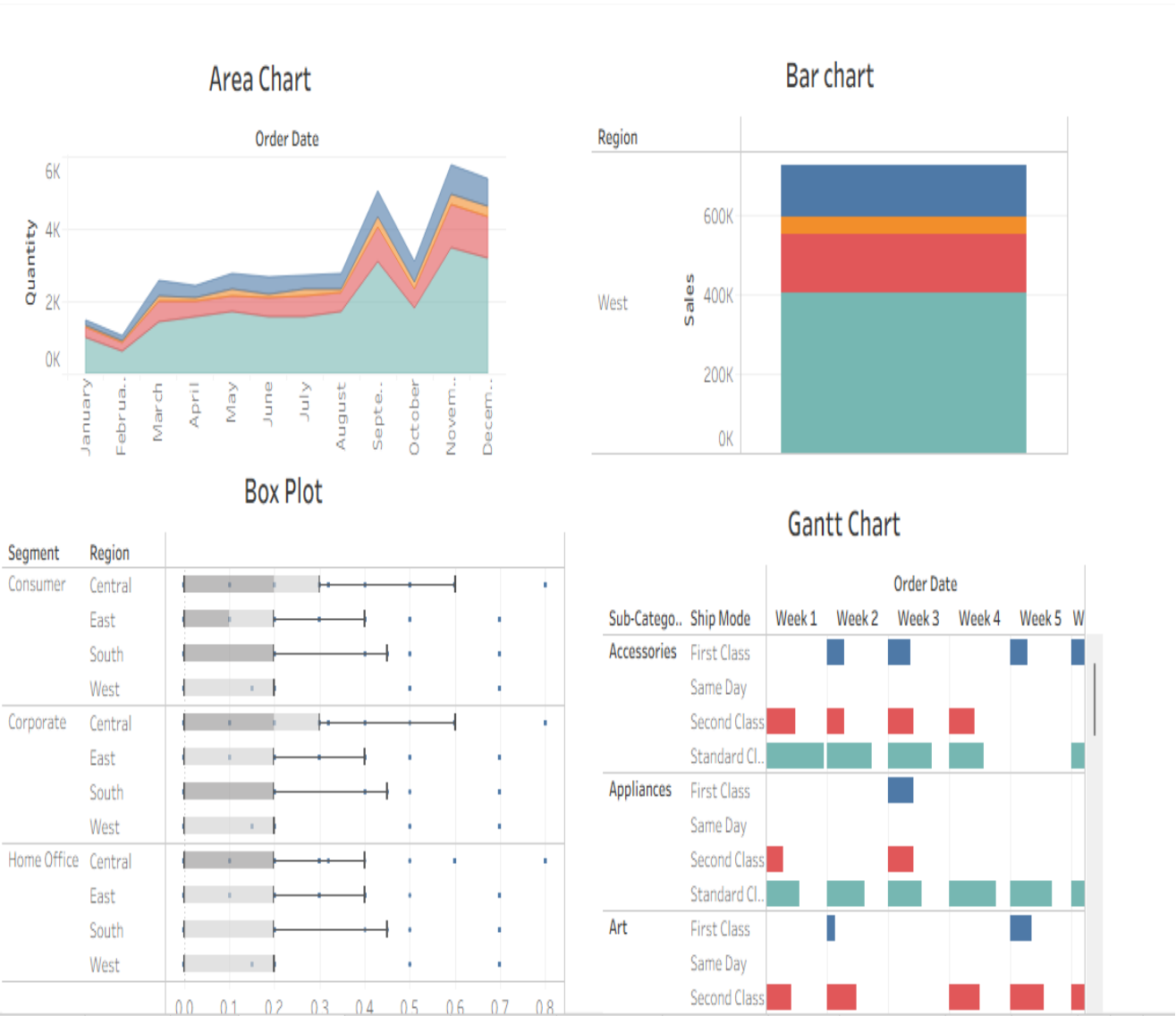
9. Key Insight: Map shows which states have highest and lowest sales. Easy to see region-wise performance.

Output:

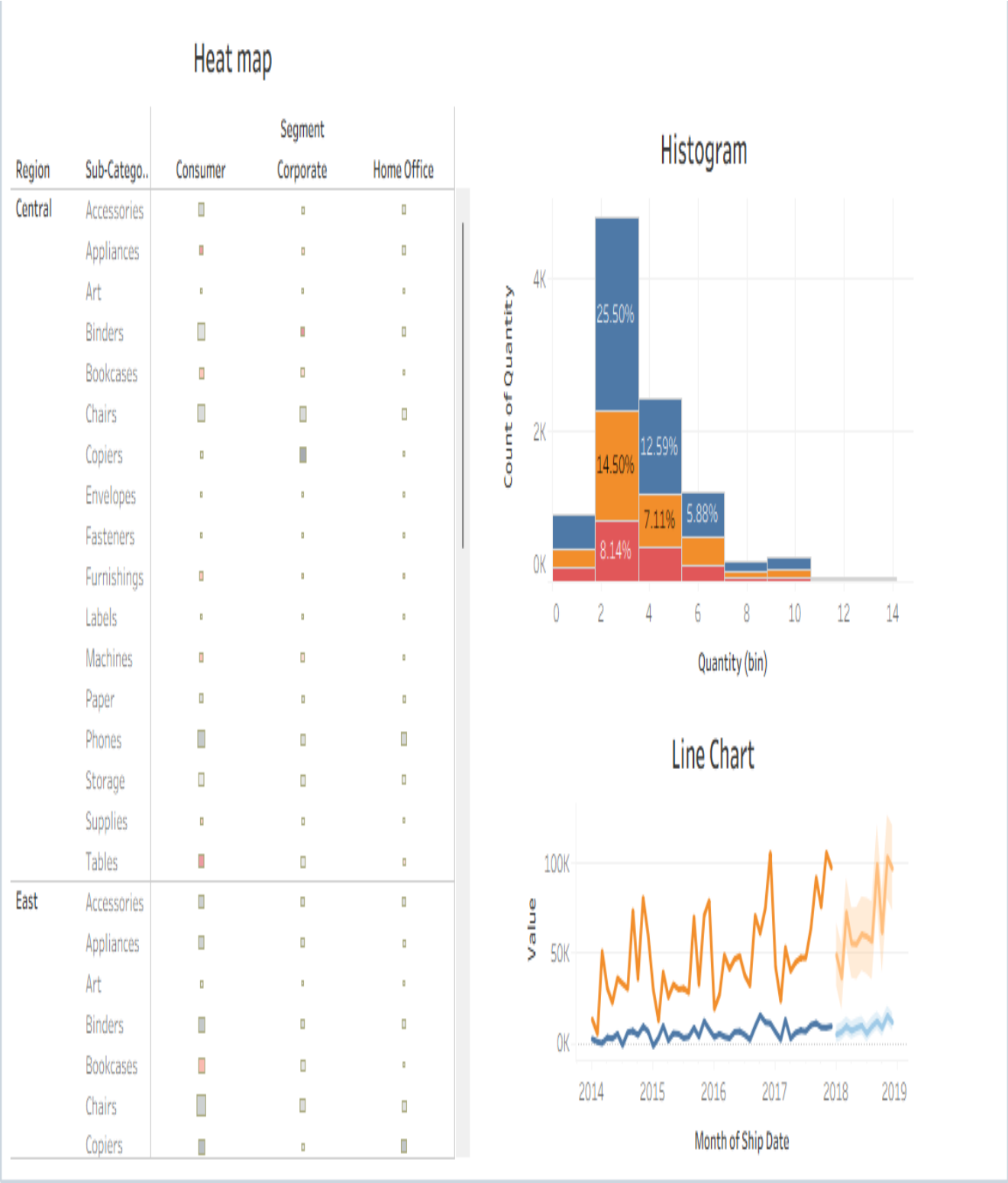


DASHBOARDS:

Dashboard-01:



Dashboards-02:



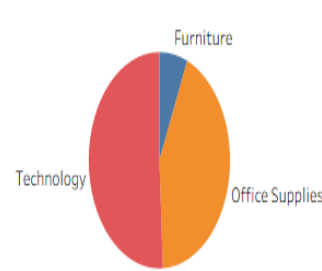


Dashboard-03:

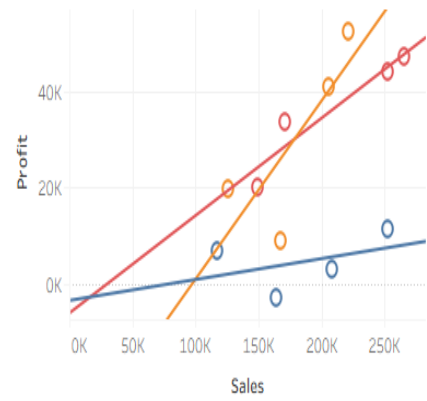
Packed Bubble Chart



Pie Chart



Scatter plot

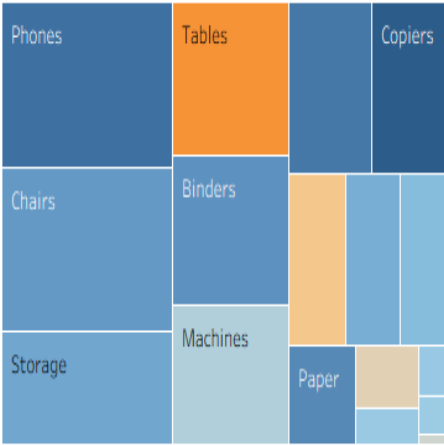


Dashboard-04:

Text Table with Percentage

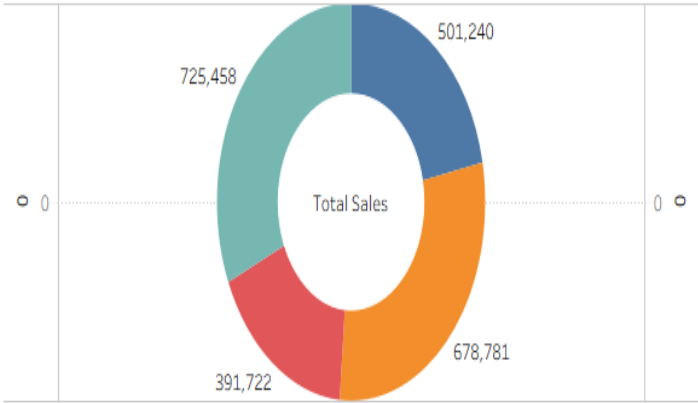
| Region  | Sub-Catego.. | Order Date |        |        |        |
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| Central | Accessories  | 4.27%      | 7.58%  | 7.33%  | 7.42%  |
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|         | Envelopes    | 1.54%      | 0.85%  | 0.66%  | 0.81%  |
|         | Fasteners    | 0.12%      | 0.09%  | 0.17%  | 0.22%  |
|         | Examinations | 2.44%      | 2.46%  | 2.47%  | 2.45%  |

Tree Map



Dashboard-05:

Dount Chart



Map Chart

